

Day 9 — Cross-cultural Calendars

Module: Moon Phases and Tithi · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: compare with hijri (islamic), chinese, and hebrew lunar calendars.

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See lesson-plan.md for any day-specific materials)

Vocabulary introduced today

- *candra* (IAST: candra; lit. “Moon”)

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **Indian lunisolar.** Lunar months (~29.5d) + adhika māsa intercalation to track the solar year. Months named for the *nakṣatra* near the full moon.
2. **Islamic Hijri.** Strictly lunar — 12 lunar months, no intercalation. ~354 days/year, drifts ~11 days/year relative to solar calendar. Reasons: religious lunar observation, no agricultural anchoring.
3. **Chinese.** Lunisolar like Indian, with intercalation. 60-year cycle combining 12 animals × 10 stems. Confucian calendar reform 104 BCE refined it.
4. **Hebrew.** Lunisolar with intercalation (Adar I/II in leap years). 19-year Metonic cycle.
5. **Roman / Gregorian.** Purely solar (since Julian 45 BCE, refined 1582 CE). Months disconnected from lunar cycle. Most universal calendar today.
6. **Source comparison.** Show a 2026 calendar in 4 traditions side-by-side. “*Which dates align? Which don’t? Why?*”

Activity (15 min)

A scaled version of *10-Day Moon Observation Diary* (see [activities/](#)). Today's slice: focus on the part of the activity that reinforces Cross-cultural Calendars.

Wrap-up (5 min)

- One-sentence exit ticket: “*What’s one thing you learned today?*”
- Preview tomorrow.

Homework

(See `homework.md` for today’s task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 *pañcabhūta*).